

Elbow Method in K-Means Clustering

The **Elbow Method** is used to determine the **optimal number of clusters (K)** for a K-Means clustering problem.

Why Do We Need It?

Before running K-Means, you must specify the number of clusters (**K**).

- If **K is too small**, different groups get merged together.
- If **K is too large**, you create unnecessary clusters and overfit the data.

The Elbow Method helps find a balance.

Step 1: Run K-Means for Different K Values

Run K-Means multiple times:

K = 1
K = 2
K = 3
K = 4
K = 5
...

For each K, calculate the **WCSS (Within-Cluster Sum of Squares)**.

WCSS is:

$$\sum_{i=1}^K \sum_{x \in C_i} \|x - \mu_i\|^2$$

A lower WCSS means points are closer to their cluster centroids.

Step 2: Create a Table

Suppose we get:

K WCSS

1 1000
2 500
3 250
4 200

K WCSS

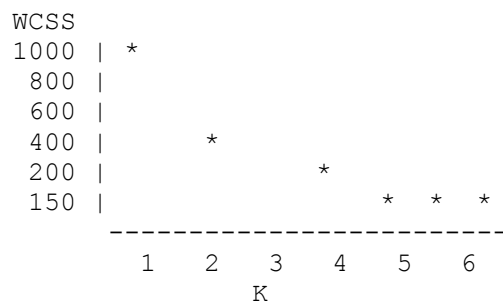
5 180

6 170

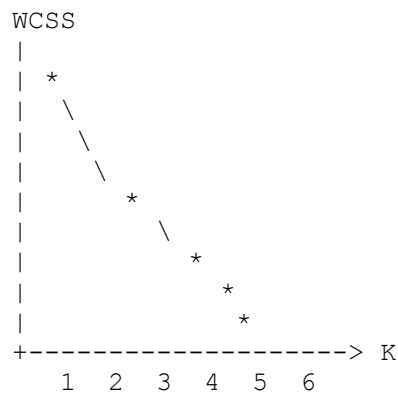
Notice:

- Moving from $K=1$ to $K=2$ gives a huge improvement.
- Moving from $K=2$ to $K=3$ also gives a big improvement.
- After $K=3$, improvements become much smaller.

Step 3: Plot K vs WCSS



A more realistic sketch:



The curve bends sharply and then starts flattening.

That bend is called the **Elbow Point**.

Step 4: Select the Elbow

In our example:

$K = 3$

is the elbow because:

- Before 3 → WCSS drops significantly.
- After 3 → WCSS decreases only slightly.

So **3 clusters** provide a good trade-off between simplicity and accuracy.

Real-Life Example: Customer Segmentation

Imagine a retailer wants to group customers.

K = 2

Budget Customers
Premium Customers

Too broad.

K = 3

Budget Customers
Regular Customers
Premium Customers

Makes sense.

K = 10

You may end up with many tiny groups that are hard to interpret and don't add much business value.

The Elbow Method helps choose a practical K.

Interview Answer (Short Version)

The Elbow Method is used to determine the optimal number of clusters in K-Means. We run K-Means for different K values, calculate WCSS for each K, and plot K versus WCSS. The point where the curve starts flattening, forming an "elbow," is chosen as the optimal number of clusters because adding more clusters beyond that point provides only marginal improvement.

Advantages

- Simple and intuitive.
- Easy to visualize.

- Widely used with K-Means.

Limitations

- The elbow is not always clearly visible.
- Different people may choose different elbow points.
- Works best when clusters are well separated.

For exams and interviews, remember:

"Elbow Method = Plot K vs WCSS and choose the K at which WCSS reduction starts slowing significantly."